29. 投 n 次停止，即前 (n-1) 次中出现一次；

$\therefore P(n) = (n-1)(1-p)^{n-2} \cdot p \cdot p$

$\therefore \sum_{n=1}^{\infty} nP(n) = p^2 \sum_{n=1}^{\infty} n(n-1)(1-p)^{n-2}$ ；设 $(1-p)=k$

$= p^2 \sum_{n=1}^{\infty} n(n-1)k^{n-2}$

$= p^2 \sum_{n=1}^{\infty} (k^n)''$

$= p^2 \cdot \left(\frac{k}{1-k}\right)''$

$= p^2 \cdot \left(\frac{1}{1-k}\right)'' = p^2 \cdot (-1) \cdot (-2) \cdot (1-k)^{-3}$

$= \frac{2}{p}$

30.

E(X|Y)	1	2	3
1	$\frac{2}{7}$	$\frac{1}{3}$	$\frac{1}{5}$
2	$\frac{4}{7}$	$\frac{7}{15}$	$\frac{2}{5}$
3	$\frac{1}{7}$	$\frac{1}{5}$	$\frac{2}{5}$

；

$\therefore E(X|Y=1) = \frac{13}{7}$

$E(X|Y=2) = \frac{22}{15}$ ；

$E(X|Y=3) = \frac{11}{5}$

∴有

E(X|Y)	$\frac{13}{7}$	$\frac{22}{15}$	$\frac{11}{5}$
P	$\frac{7}{27}$	$\frac{15}{27}$	$\frac{5}{27}$

$\therefore E(X) = P(Y=1) \cdot \left(\frac{2}{7} + \frac{8}{7} + \frac{3}{7}\right) + P(Y=2) \cdot \left(\frac{1}{3} + \frac{14}{15} + \frac{3}{15}\right) + P(Y=3) \cdot \left(\frac{1}{5} + \frac{4}{5} + \frac{6}{5}\right)$

$= \frac{52}{27}$

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31. 可视为随机游走

$\because$ Y_2 : -2 0 2

Y_3 : -3 q^3 0 0

-1 $3q^2p$ $2pq^2$ 0

1 0 $2pq^2$ $3qp^2$

3 0 0 p^3

$\therefore$ $E(Y_3 \mid Y_2)$: $-p-3q$ $(p-q)$ $3p+q$

p q^2 $1-q^2-p^2$ p^2

$\therefore E(Y_{n+1} \mid Y_n) = Y_n + E(Y_1) = Y_n + (p-q)$

32.

(a) 即前 $(n-1)$ 件为 1 类 2 等品.

$\therefore S = (n-1) \cdot \frac{p_1}{p_1+p_2}$

(b) $\because$ n 次生产后修概率

$P = (1-p_3)^{n-1} \cdot p_3$

$\therefore S = \sum_{n=1}^{\infty} (n-1) \cdot \frac{p_1}{p_1+p_2} \cdot (1-p_3)^{n-1} \cdot p_3$

$= (1-p_3) \cdot \frac{p_1 p_3}{p_1+p_2} \cdot \sum_{n=1}^{\infty} (n-1) \cdot (1-p_3)^{n-2}$

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设 $(1-p_3)=k$；

$$\sum_{n=1}^{\infty}(n-1)k^{n-2}=\sum_{n=1}^{\infty}(k^{n-1})'=\left(\frac{1}{1-k}\right)'=-1\cdot(1-k)^{-2}(-1)=\frac{1}{(1-k)^2}$$

$$\therefore S=(1-p_3)\cdot\frac{p_1p_3}{p_1+p_2}\cdot\frac{1}{p_3^2}=(1-p_3)\cdot\frac{p_1}{p_3(p_1+p_2)}=\frac{p_1}{p_3}$$

35.

X	0	1	2
P	$\frac{3}{4}\cdot\frac{2}{7}+\frac{1}{4}\cdot\frac{4}{7}$	$\frac{2}{7}$	$\frac{1}{2}\cdot\frac{3}{7}+\frac{1}{2}\cdot\frac{4}{7}$

即

L \ X	0	1	2
2	$\frac{3}{7}$	$\frac{1}{7}$	0
3	0	0	$\frac{1}{7}$
4	$\frac{1}{7}$	0	$\frac{1}{7}$

$E(L|X=0)=\frac{5}{2}$；

$E(L|X=1)=2$

$E(L|X=2)=\frac{7}{2}$

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36.

$D(X|Y) = E[(X - E(X|Y))^2 | Y]$

又 $\because DX = E(X^2) - E(X)^2$

且 $D[E(X|Y)] = E[E(X|Y)^2] - E(E(X|Y))^2$

$\therefore E[D(X|Y)] + D[E(X|Y)]$

$= E(X - E(X|Y))^2 + E[E(X|Y)^2] - E(E(X|Y))^2$

$= EX^2 + 2E[E(X|Y)]^2 - 2E(X\ E(X|Y)) - E(X)^2$

$= DX$.

37. (a) $P = p^4 + 3p^3q + p^2q^2 = p^2(2-p)$

(b) $P = q^3 + q^2 \cdot p \cdot q = (1-p)^3 \cdot (1+p)$

(c) $P = 1 - 2pq - 2p^2q^2$

(d) $P = 1 - p^3 - q^3$

39. (a) $P(X_{n+1} = i_{n+1} | X_n = i_n, \cdots X_0 = i_0)$

$= P(X_{n+1} - X_n = i_{n+1} - i_n)$

与之前无关.

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(b) $P(X_{n+1}=i_{n+1}, X_{n-1}=i_{n-1} \mid X_n=i_n)$

$= \frac{P(X_{n+1}=i_{n+1}, X_n=i_n, X_{n-1}=i_{n-1})}{P(X_n=i_n)} = \frac{P(X_{n-1}=i_{n-1}, X_n=i_n)}{P(X_n=i_n)} P(X_{n+1}=i_{n+1} \mid X_{n-1}=i_{n-1})$

$= P(X_{n+1}=i_{n+1} \mid X_n=i_n) \cdot P(X_{n-1}=i_{n-1} \mid X_n=i_n)$

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补充：

设抛数为随机变量 X，● 第 k 次为正面记 $Y_k=1$，否则 $Y_k=0$

$\therefore E(X) = E(X \mid Y_1=1) P(Y_1=1) + E(X \mid Y_1=0) P(Y_1=0)$

$= \frac{1}{2} E(X \mid Y_1=1) + \frac{1}{2}(1+E(X))$

$\therefore E(X) - E(X \mid Y_1=1) = 1$

且 $E(X \mid Y_1=1) = E(X \mid Y_1=1, Y_2=1) ● \cdot \frac{1}{2} + E(X \mid Y_1=1, Y_2=0) \cdot \frac{1}{2}$

$= \frac{1}{2} \cdot 2 + \frac{1}{2} \cdot (2+E(X))$

$\therefore E(X \mid Y_1=1) = ● \frac{1}{2} E(X) + 2$

$\therefore \frac{1}{2} E(X) - 2 = 1; \quad E(X) = 6$